



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 6 • STANDARD 6.NOS.1

Divide Mixed Numbers

Lesson 6-10 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can divide mixed numbers by first changing them to improper fractions.

Language Objective: I can explain my steps using the words mixed number, improper fraction, convert, reciprocal, and simplify.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Mixed Number

Improper Fraction

Convert

Simplify

Reciprocal



Tap any word to see its meaning and a picture.

1 A number with a whole number and a fraction, like $2\frac{1}{3}$, is a

2 A fraction whose numerator is greater than or equal to its denominator, like $\frac{7}{3}$, is an

3 Changing a mixed number into an improper fraction is to

 it.

4 Writing a fraction with smaller numbers that names the same amount, like $\frac{4}{8} = \frac{1}{2}$, is to

 it.

5 To divide by a fraction, I multiply by its

Write About the Math

How does dividing mixed numbers help the detective?

¿Cómo ayuda dividir números mixtos a la detective?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Mixed Number**
Número mixto

☐ **Improper Fraction**
Fracción impropia

☐ **Convert**
Convertir

☐ **Simplify**
Simplificar

☐ **Reciprocal**
Recíproco

Model: A mixed number must first become an improper fraction so I can flip and multiply. — Students explain that the mixed number must be converted to an improper fraction ($7/2$) before applying Keep, Change, Flip.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- The detective can cut ____ pieces because $5 \frac{1}{4} \div \frac{3}{4} = \frac{21}{4} \times \frac{4}{3} = \frac{84}{12} = \underline{\hspace{1cm}}$.
- My answer is ____ because ____.

Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is ____.
Mi afirmación es ____.
- My evidence is ____, so ____.
Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Mixed Number
2. **Notes 2:** Improper Fraction
3. **Notes 3:** Convert
4. **Notes 4:** Simplify
5. **Notes 5:** Reciprocal
6. **Watch for this mistake:** *A common mistake in Divide Mixed Numbers is dividing the whole-number parts and fraction parts separately instead of converting first — for example, treating $4\frac{1}{2} \div 1\frac{1}{2}$ as $(4 \div 1) + (1/2 \div 1/2) = 5$. Always convert each mixed number to one improper fraction first, then divide using Keep, Change, Flip: $4\frac{1}{2} \div 1\frac{1}{2} = 9/2 \div 3/2 = 9/2 \times 2/3 = 3$.*
7. **Try It 1:** B. $2 - 3 \div 1\frac{1}{2} = 3 \div 3/2 = 3 \times 2/3 = 6/3 = 2$ sections.
8. **Try It 2:** B. $3 - 7\frac{1}{2} = 15/2$ and $2\frac{1}{2} = 5/2$. So $15/2 \div 5/2 = 15/2 \times 2/5 = 30/10 = 3$ doorways.